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Studierichting: Informatica
Oefeningenreeks 5G nr. 1.13

$f(x) = \sqrt{x}\sqrt[4]{1-x}$ op $[0, 1]$

x	0		$\frac{2}{3}$		1
$f(x)$	0		$+$		0
$f'(x)$	$+$		0		$-$
$f''(x)$	$-$		$-$		$-$
$f(x)$	0	$\nearrow$	$M\left(\frac{\sqrt{2}}{3^{\frac{3}{4}}}\right)$	$\searrow$	0
		$\smile$		$\frown$	

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$$V = \pi \int_0^1 \left(\sqrt{x}\sqrt[4]{1-x}\right)^2 dx$$
$$= \pi \int_0^1 x\sqrt{1-x}dx$$
$$t = 1 - x\,(x = 1 - t,\ dx = -dt)$$
$$= \pi \int_1^0 (1-t)\sqrt{t}(-dt)$$
$$= \pi \int_0^1 (1-t)t^{1/2}dt$$
$$= \pi \int_0^1 \left(t^{1/2} - t^{3/2}\right) dt$$
$$= \pi \left[\frac{2}{3}t^{3/2} - \frac{2}{5}t^{5/2}\right]_0^1$$
$$= \pi \left(\frac{2}{3} - \frac{2}{5}\right)$$
$$= \frac{4\pi}{15}$$

References